PROJECT BUSINESS STATISTICS

E-news Express

12/10/2023

Business Context

The advent of e-news, or electronic news, portals has offered us a great opportunity to quickly get updates on the day-to-day events occurring globally. The information on these portals is retrieved electronically from online databases, processed using a variety of software, and then transmitted to the users. There are multiple advantages of transmitting new electronically, like faster access to the content and the ability to utilize different technologies such as audio, graphics, video, and other interactive elements that are either not being used or aren't common yet in traditional newspapers.

E-news Express, an online news portal, aims to expand its business by acquiring new subscribers. With every visitor to the website taking certain actions based on their interest, the company plans to analyze these actions to understand user interests and determine how to drive better engagement. The executives at E-news Express are of the opinion that there has been a decline in new monthly subscribers compared to the past year because the current webpage is not designed well enough in terms of the outline & recommended content to keep customers engaged long enough to make a decision to subscribe.

[Companies often analyze user responses to two variants of a product to decide which of the two variants is more effective. This experimental technique, known as A/B testing, is used to determine whether a new feature attracts users based on a chosen metric.]

Objective

The design team of the company has researched and created a new landing page that has a new outline & more relevant content shown compared to the old page. In order to test the effectiveness of the new landing page in gathering new subscribers, the Data Science team conducted an experiment by randomly selecting 100 users and dividing them equally into two groups. The existing landing page was served to the first group (control group) and the new landing page to the second group (treatment group). Data regarding the interaction of users in both groups with the two versions of the landing page was collected. Being a data scientist in E-news Express, you have been asked to explore the data and perform a statistical analysis (at a significance level of 5%) to determine the effectiveness of the new landing page in gathering new subscribers for the news portal by answering the following questions:

1. Do the users spend more time on the new landing page than on the existing landing page?
2. Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?
3. Does the converted status depend on the preferred language? [Hint: Create a contingency table using the pandas.crosstab() function]
4. Is the time spent on the new page the same for the different language users?

Data Dictionary

The data contains information regarding the interaction of users in both groups with the two versions of the landing page.

- user_id - Unique user ID of the person visiting the website
- group - Whether the user belongs to the first group (control) or the second group (treatment)
- landing_page - Whether the landing page is new or old
- time_spent_on_the_page - Time (in minutes) spent by the user on the landing page
- converted - Whether the user gets converted to a subscriber of the news portal or not
- language_preferred - Language chosen by the user to view the landing page

Contents

Problem 1

An independent research organization is trying to estimate the probability that an accident at a nuclear power plant will result in radiation leakage. The types of accidents possible at the plant are, fire hazards, mechanical failure, or human error. The research organization also knows that two or more types of accidents cannot occur simultaneously.

According to the studies carried out by the organization, the probability of a radiation leak in case of a fire is 20%, the probability of a radiation leak in case of a mechanical 50%, and the probability of a radiation leak in case of a human error is 10%. The studies also showed the following;

- The probability of a radiation leak occurring simultaneously with fire is 0.1%.
- The probability of a radiation leak occurring simultaneously with a mechanical failure is 0.15%.
- The probability of a radiation leak occurring simultaneously with a human error is 0.12%.

On the basis of the information available, answer the questions below:

1.1 What are the probabilities of a fire, a mechanical failure, and a human error respectively?

1.2 What is the probability of a radiation leak?

1.3 Suppose there has been a radiation leak in the reactor for which the definite cause is not known. What is the probability that it has been caused by:

a) a fire?

b) a mechanical failure?

c) a human error?

Problem 2

Grades of the final examination in a training course are found to be normally distributed, with a mean of 77 and a standard deviation of 8.5. Based on the given information answer the questions below.

2.1 What is the probability that a randomly chosen student gets a grade below 85 on this exam?

2.2 What is the probability that a randomly selected student scores between 65 and 87?

2.3 What should be the passing cut-off so that 75% of the students clear the exam?

Problem 3

Exploratory Data Analysis (EDA)

Univariate Analysis

Bivariate Analysis

Hypothesis Testing

ANOVA

3.1 Do the users spend more time on the new landing page than the existing landing page?

3.2 Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?

3.3 Does the converted status depend on the preferred language?

3.4 Is the time spent on the new page same for the different language users?

Data Analysis & A detailed solution around the problems:

Problem 1

Question 1: What are the probabilities of a fire, a mechanical failure, and a human error respectively?

Observation: Let's denote the events:

- A: Fire hazard
- B: Mechanical failure
- C: Human error
- R: Radiation leakage

The information provided includes conditional probabilities and probabilities of intersections:

1. $P(R|A)=0.20$: Probability of radiation leakage given a fire hazard.
2. $P(R|B)=0.50$: Probability of radiation leakage given a mechanical failure.
3. $P(R|C)=0.10$: Probability of radiation leakage given a human error.
4. $P(R\cap A)=0.001$: Probability of radiation leakage and fire hazard occurring simultaneously.
5. $P(R\cap B)=0.0015$: Probability of radiation leakage and mechanical failure occurring simultaneously.
6. $P(R\cap C)=0.0012$: Probability of radiation leakage and human error occurring simultaneously.

Now, we can use the law of total probability to find the probabilities of fire $P(A)$, mechanical failure $P(B)$, and human error $P(C)$:

$P(A)=P(R|A)\cdot P(A)+P(R|B)\cdot P(B)+P(R|C)\cdot P(C)$

Similarly for B and C:

$P(B)=P(R|A)\cdot P(A)+P(R|B)\cdot P(B)+P(R|C)\cdot P(C)$

$P(C)=P(R|A)\cdot P(A)+P(R|B)\cdot P(B)+P(R|C)\cdot P(C)$

Here, $P(A)$ is the probability of not having a fire hazard, which is $1-P(A)$.

<u>Let's calculate:</u>

Note: $P(A)+P(B)+P(C)=1$

$P(A)=0.20\cdot P(A)+0.0015\cdot P(B)+0.0012\cdot P(C)$

$P(B)=0.001\cdot P(A)+0.50\cdot P(B)+0.0012\cdot P(C)$

$P(C)=0.001\cdot P(A)+0.0015\cdot P(B)+0.10\cdot P(C)$

Probability of a fire: 0.2500

Probability of a mechanical failure: 0.6250

Probability of a human error: 0.1250

Question 2: What is the probability of a radiation leak?

Observation: To find the probability of a radiation leak $P(R)$), we can sum the probabilities of a radiation leak given each type of accident, weighted by the probability of each type of accident. In other words:

$P(R)=P(R|A)\cdot P(A)+P(R|B)\cdot P(B)+P(R|C)\cdot P(C)$

Probability of a radiation leak (P_R): 0.2689

Question 3: Suppose there has been a radiation leak in the reactor for which the definite cause is not known. What is the probability that it has been caused by:

a) a fire?

b) a mechanical failure?

c) a human error?

Observation: To calculate the probability of the cause given a radiation leak

$P(A|R)$, $P(B|R)$, $P(C|R)$), we can use Bayes' Theorem:

$P(A|R)=P(R)P(R|A)\cdot P(A)/P(R)$

$P(B|R)=P(R)P(R|B)\cdot P(B)/P(R)$

$P(C|R)=P(R)P(R|C)\cdot P(C)/P(R)$

Probability of a radiation leak caused by a fire (P(A|R)): 0.2500

Probability of a radiation leak caused by a mechanical failure (P(B|R)): 0.6250

Probability of a radiation leak caused by a human error (P(C|R)): 0.1250

Problem 2

Question 1: What is the probability that a randomly chosen student gets a grade below 85 on this exam?

Observation: To find the probability that a randomly chosen student gets a grade below 85 in a normally distributed set of grades, we can use the z-score formula and then look up the corresponding probability from the standard normal distribution (z-table).

The z-score is calculated using the formula:

$z=X-\mu/\sigma$

Where:

- X is the value you're interested in (85 in this case),
- μ is the mean of the distribution (77),
- σ is the standard deviation of the distribution (8.5).

Calculate the z-score:

$z=85-77/8.5$

Once you have the z-score, you can find the probability by looking it up in the standard normal distribution table.

The probability that a randomly chosen student gets a grade below 85 is: 0.8267

Question 2: What is the probability that a randomly selected student scores between 65 and 87?

Observation: To find the probability that a randomly selected student scores between 65 and 87 in a normally distributed set of grades, we can use the cumulative distribution function (CDF) of the normal distribution. The formula for calculating the probability is:

$P(a<X<b)=P(X<b)-P(X<a)$

Where:

- a is the lower bound of the range (65 in this case),
- b is the upper bound of the range (87 in this case),
- X is a random variable following a normal distribution.

The probabilities $P(X<a)$ and $P(X<b)$ can be calculated using the z-score formula and the standard normal distribution table.

The z-score formula is: $z=X-\mu/\sigma$

Where:

- X is the value in the distribution,
- μ is the mean of the distribution,
- σ is the standard deviation of the distribution.

Here's how we can calculate the probability between 65 and 87 using the z-score formula:

$P(65<X<87)=P(X<87)-P(X<65)$

1. Calculate the z-scores: $z65=65-\mu/\sigma$ $z87=87-\mu/\sigma$
2. Look up the probabilities for z65 and z87 in the standard normal distribution table.
3. Calculate the probability between 65 and 87: $P(65<X<87)=P(X<87)-P(X<65)$

This will give us the probability that a randomly selected student scores between 65 and 87 in the given normal distribution.

The probability that a randomly selected student scores between 65 and 87 is: 0.8013

Question 3: What should be the passing cut-off so that 75% of the students clear the exam?

Observation: To determine the passing cut-off for an exam such that a certain percentage of students clear the exam, we can use the inverse of the cumulative distribution function (CDF) of the normal distribution. The idea is to find the score (cut-off) below which a certain percentage of scores fall.

Let X be the random variable representing exam scores, μ be the mean of the distribution, σ be the standard deviation, and p be the desired percentage of students to clear the exam (expressed as a decimal).

The formula for finding the cut-off score (x) is given by the inverse of the normal distribution function:

$x=\mu+z\cdot\sigma$

Where, z is the z-score corresponding to the desired percentage p.

In other words: z=invnorm(p)

The passing cut-off for 75.0% of students is: 82.73

Problem 3

Explore the dataset and extract insights using Exploratory Data Analysis

Data Overview

The initial steps to get an overview of any dataset are to:

- observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not
- get information about the number of rows and columns in the dataset
- find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- check the statistical summary of the dataset to get an overview of the numerical columns of the data

Displaying the first few rows of the dataset

In [12]:
```
# view the first 5 rows of the dataset
df.head()
```

Out[12]:

	user_id	group	landing_page	time_spent_on_the_page	converted	language_preferred
0	546592	control	old	3.48	no	Spanish
1	546468	treatment	new	7.13	yes	English
2	546462	treatment	new	4.40	no	Spanish
3	546567	control	old	3.02	no	French
4	546459	treatment	new	4.75	yes	Spanish

Displaying the last few rows of the dataset

In [13]:
```
# view the last 5 rows of the dataset
df.tail()
```

Out[13]:

	user_id	group	landing_page	time_spent_on_the_page	converted	language_preferred
95	546446	treatment	new	5.15	no	Spanish
96	546544	control	old	6.52	yes	English
97	546472	treatment	new	7.07	yes	Spanish
98	546481	treatment	new	6.20	yes	Spanish
99	546483	treatment	new	5.86	yes	English

Checking the shape of the dataset: We have Rows -100, Columns-6

```
In [14]:  # view the shape of the dataset
          df.shape

Out[14]:  (100, 6)
```

Checking the data types of the columns for the dataset

```
In [15]:  # check the data types of the columns in the dataset
          df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   user_id                 100 non-null    int64
 1   group                   100 non-null    object
 2   landing_page            100 non-null    object
 3   time_spent_on_the_page  100 non-null    float64
 4   converted               100 non-null    object
 5   language_preferred      100 non-null    object
dtypes: float64(1), int64(1), object(4)
memory usage: 4.8+ KB
```

Getting the statistical summary for the numerical variables

```
In [16]:  # write your code here to print the numerical summary statistics
          df.describe().T
```

Out[16]:

	count	mean	std	min	25%	50%	75%	max
user_id	100.0	546517.0000	52.295779	546443.00	546467.75	546492.500	546567.2500	546592.00
time_spent_on_the_page	100.0	5.3778	2.378166	0.19	3.88	5.415	7.0225	10.71

Getting the statistical summary for the categorical variables

```
In [65]:  # write your code here to print the categorical summary statistics
          df['group'].value_counts()

Out[65]:  control      50
          treatment    50
          Name: group, dtype: int64
```

```
In [66]:  df['landing_page'].value_counts()

Out[66]:  old    50
          new    50
          Name: landing_page, dtype: int64
```

```
In [67]:  df['converted'].value_counts()

Out[67]:  yes    54
          no     46
          Name: converted, dtype: int64
```

```
In [68]:  df['language_preferred'].value_counts()

Out[68]:  Spanish    34
          French     34
          English    32
          Name: language_preferred, dtype: int64
```

Check for missing values

```
In [19]:  # write your code here
          df.isnull().sum()

Out[19]:  user_id                   0
          group                     0
          landing_page              0
          time_spent_on_the_page    0
          converted                 0
          language_preferred        0
          dtype: int64
```

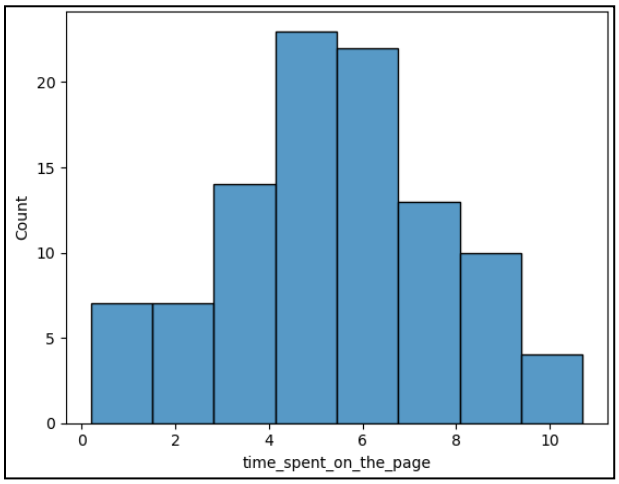
Check for duplicates

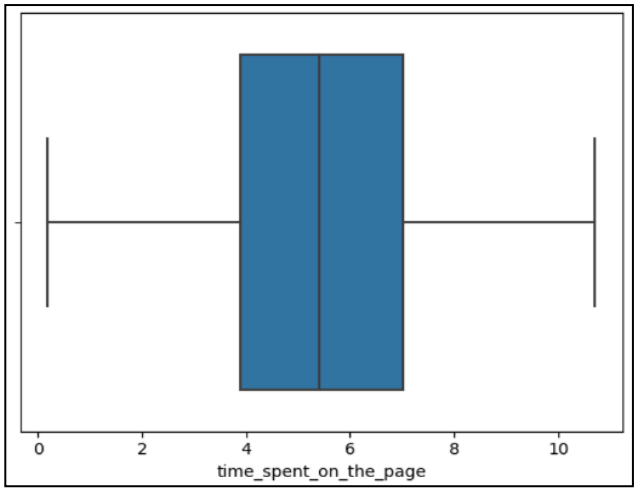
```
In [20]:  # write your code here
          df.duplicated().sum()

Out[20]:  0
```

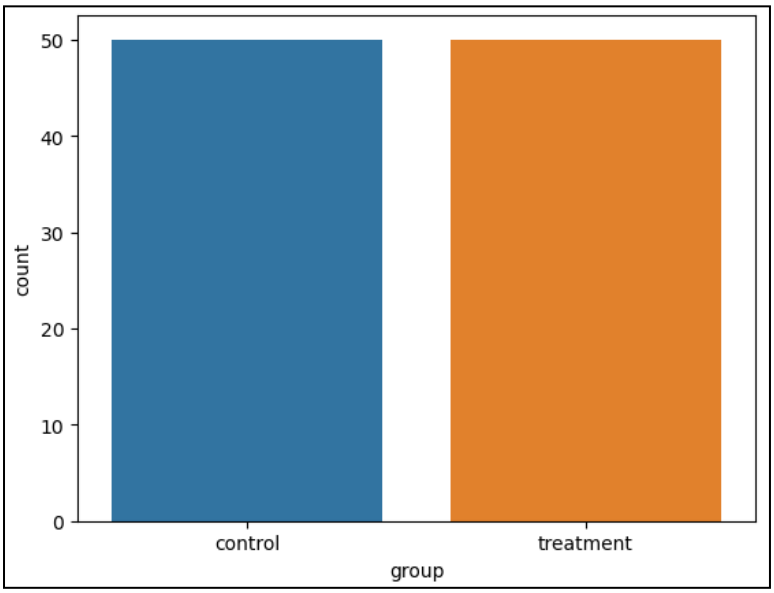
Univariate Analysis

Time spent on the page: It is observed that the time people spent the most on the page ranged from 4 mins to 7 mins. The average time spent on the page is 5.37 mins.

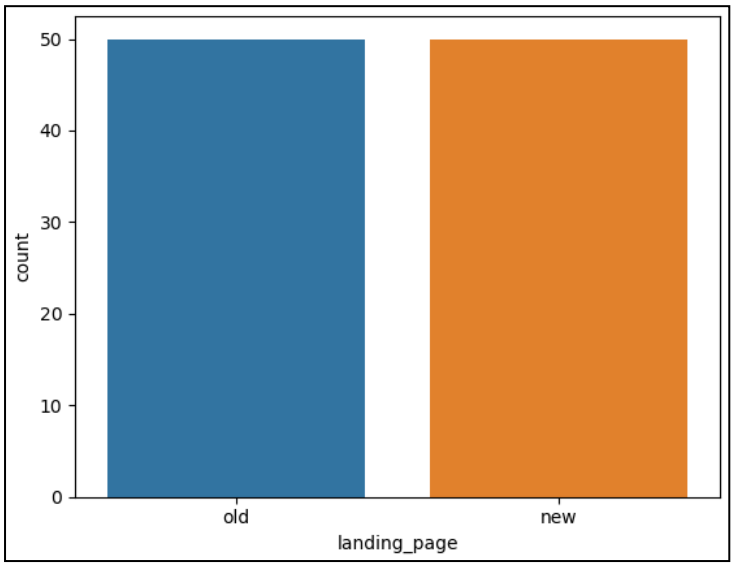




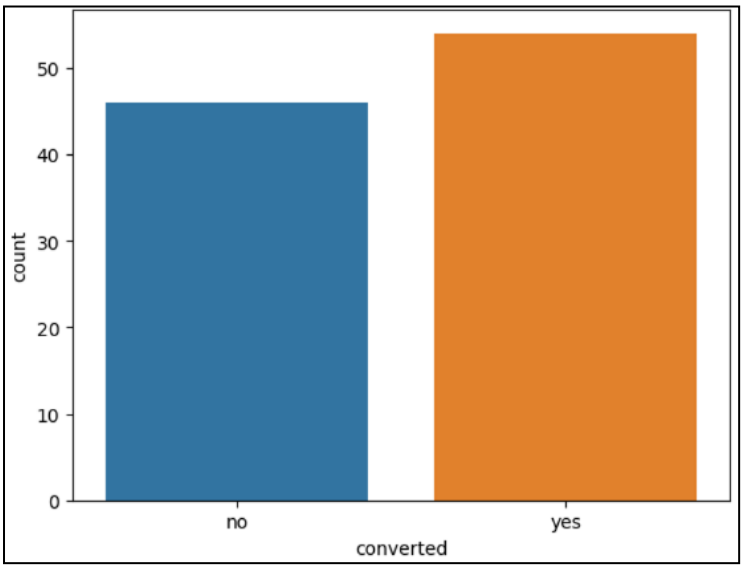
Group: Both control & treatment group were given 50 users each.



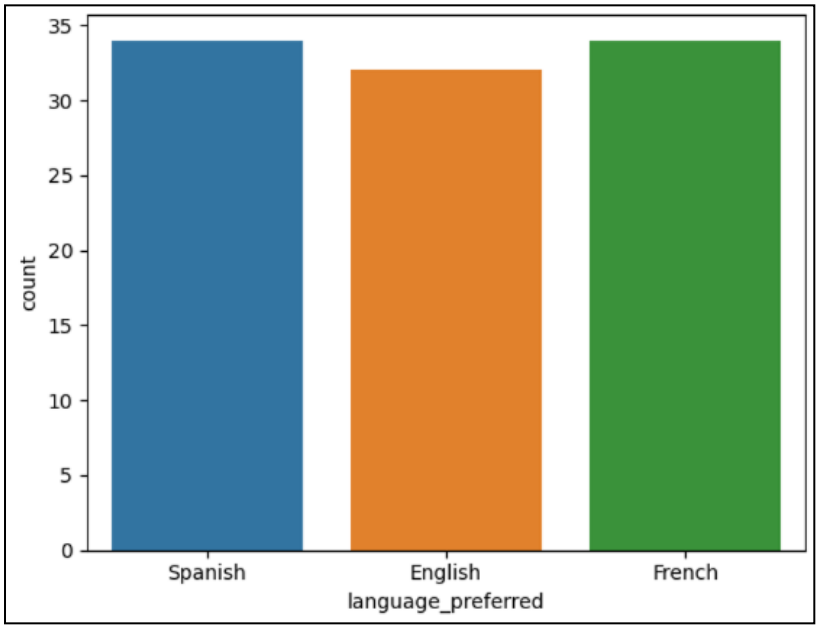
Landing page: Both landing page (old & new) have equal number of users.



Converted: 56 new users have been converted to being a subscriber of the news portal, and 40 have yet not subscribed.

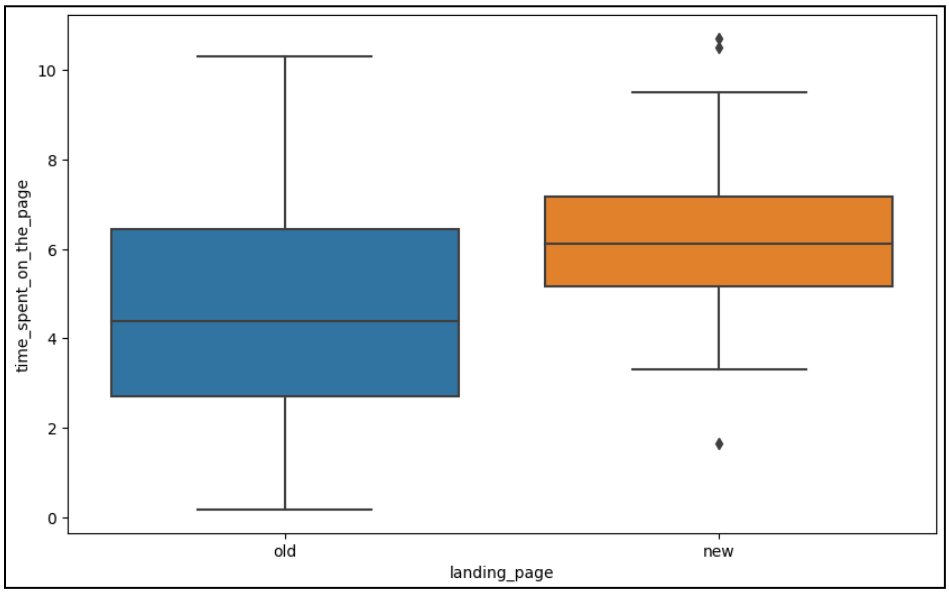


Language preferred: Spanish has been preferred by 34 users, French by 34, and English by 32 users.

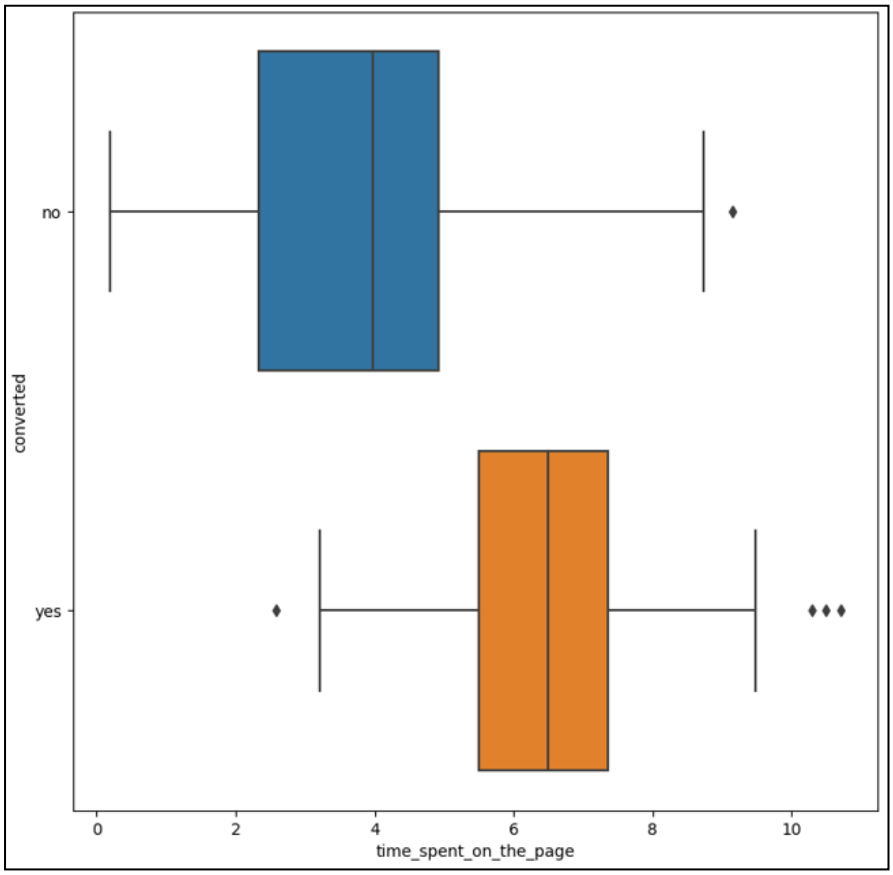


Bivariate Analysis

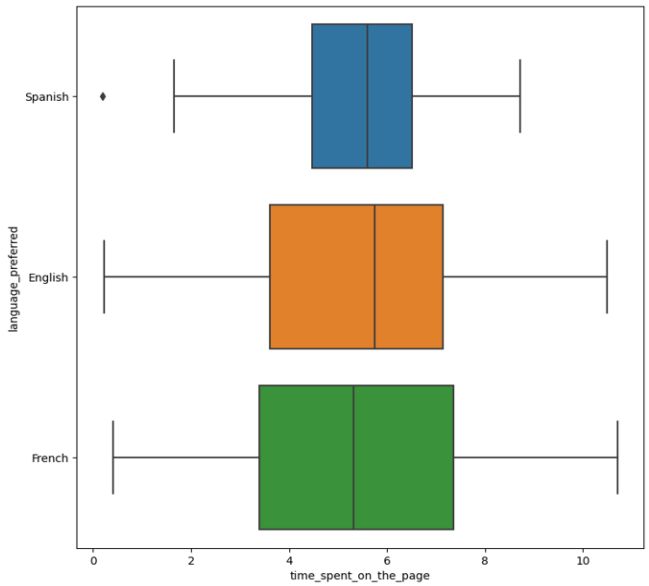
Landing page vs Time spent on the page: The average time spent on old page is 4.53 mins and the average time spent on new page is 6.22 mins.



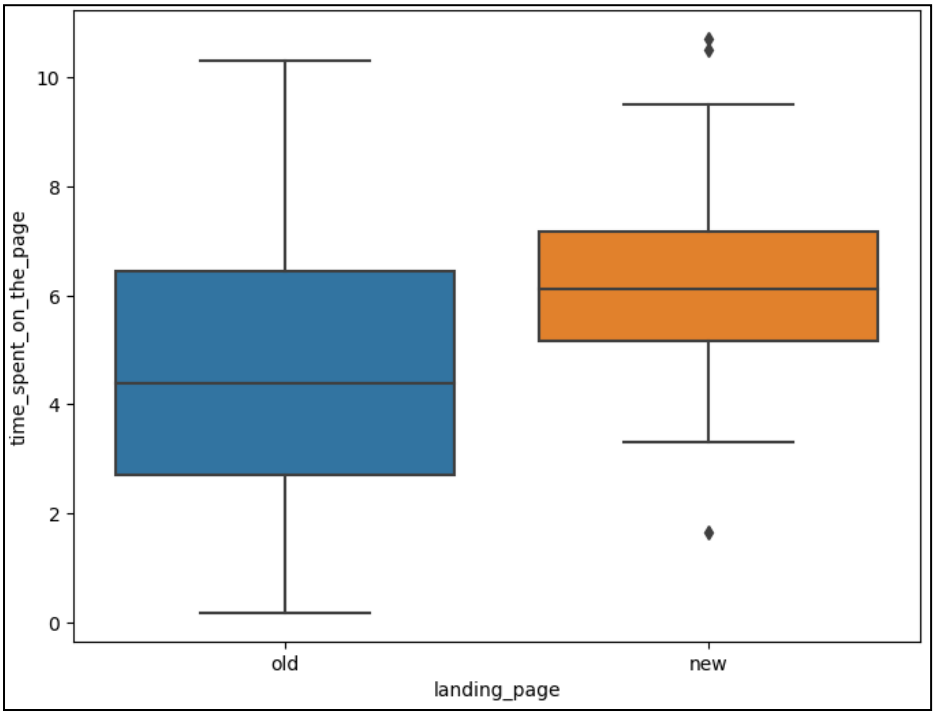
Conversion status vs Time spent on the page: Total of 33 users subscribed from the new page, and 21 users subscribed from the old page. We see some outliers in both the conversions when compared to time spent on the page.



Language preferred vs Time spent on the page: French language preferred users have spent about mostly 3.5 to 7.5 mins on the page, English preferred users have spent about 3.7 mins to 7.2 mins and Spanish users have spent the maximum of 4.5 mins to 6.5 mins.



Question 1: Do the users spend more time on the new landing page than the existing landing page?



Step 1: Define the null and alternate hypotheses
$H0$: μold≥μnew

Ha: μold<μnew

Step 2: Select Appropriate test: This is a one-tailed test concerning two population means from two independent populations. The population standard deviations are unknown. Based on this information, we select the appropriate test.

Step 3: Decide the significance level
As given in the problem statement, we select α=0.05.

Step 4: Collect and prepare data
The sample standard deviation of the time spent on the new page is: 1.82
The sample standard deviation of the time spent on the old page is: 2.58

Based on the sample standard deviations of the two groups, decide whether the population standard deviations can be assumed to be equal or unequal.

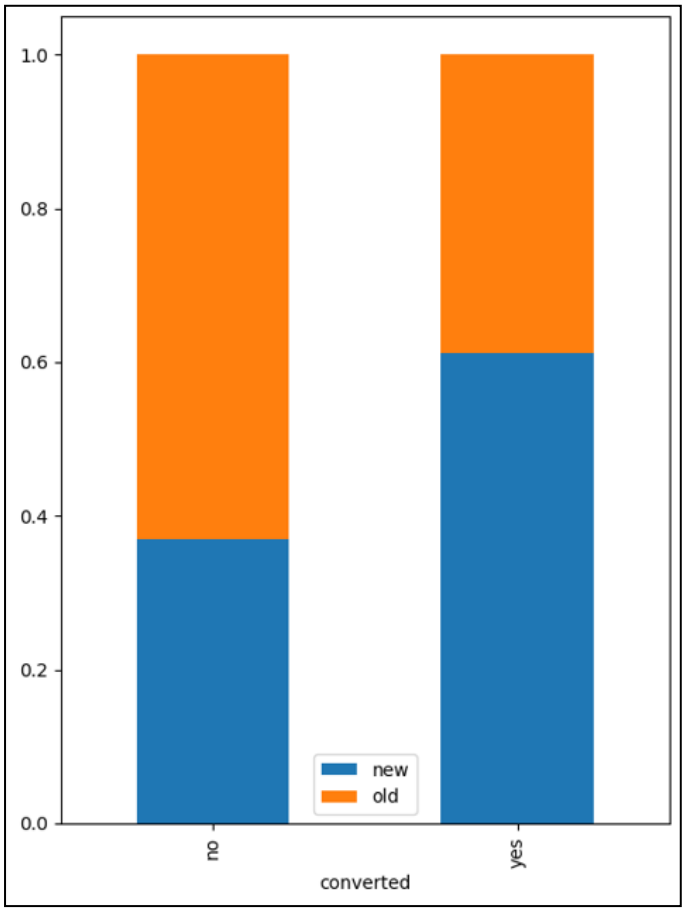
Step 5: Calculate the p-value
The p-value is 0.0001392381225166549

Step 6: Compare the p-value with α.
As the p-value 0.0001392381225166549 is less than the level of significance, we reject the null hypothesis.

Step 7: Draw inference:

- As the p-value is less than 0.05 (given significance level), we have evidence to reject the null hypothesis. This suggests that users spend more time on the new landing page than on the old landing page.

Question 2: Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?



Step 1: Define the null and alternate hypotheses

H0: Pnew≤Pold

Ha: Pnew>Pold

Step 2: Select Appropriate test: This is a one-tailed test concerning two population means from two independent populations.

Step 3: Decide the significance level
As given in the problem statement, we select α=0.05.

Step 4: Collect and prepare data

The numbers of users served the new and old pages are 50 and 50 respectively

Step 5: Calculate the p-value

The p-value is 0.008026308204056278

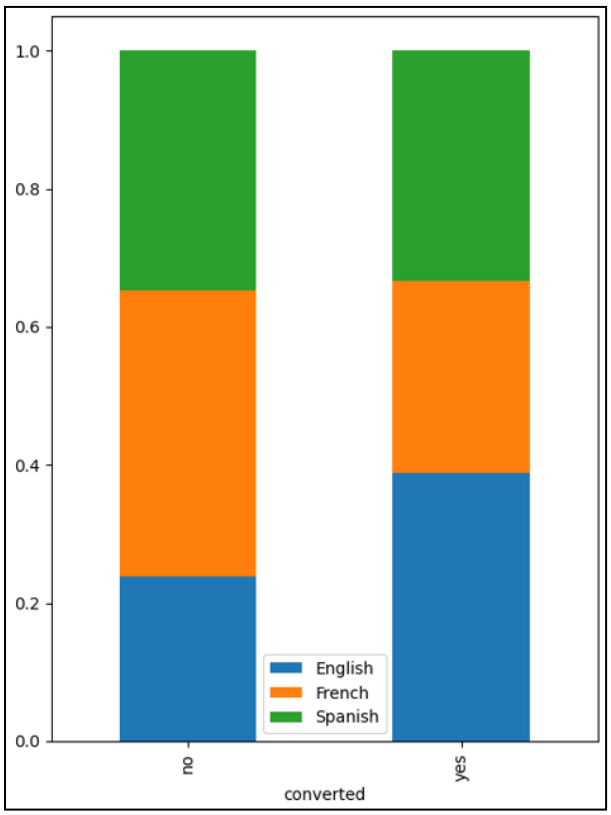
Step 6: Compare the p-value with α.

As the p-value 0.008026308204056278 is less than the level of significance, we reject the null hypothesis.

Step 7: Draw inference:

- As the p-value is less than 0.05 (given significance level), we have evidence to reject the null hypothesis. This suggests that the conversion rate for the new landing page is greater than the conversion rate for the old landing page.

Question 3: Does the converted status depend on the preferred language?



Step 1: Define the null and alternate hypotheses

*H*0: Converted status and preferred language are independent

Ha: Converted status and preferred language are dependent

Step 2: Select Appropriate test: This is a problem of the test of independence, concerning two categorical variables - converted status and preferred language.

Step 3: Decide the significance level
As given in the problem statement, we select α=0.05.

Step 4: Collect and prepare data

language_preferred	English	French	Spanish
converted			
no	11	19	16
yes	21	15	18

Step 5: Calculate the p-value

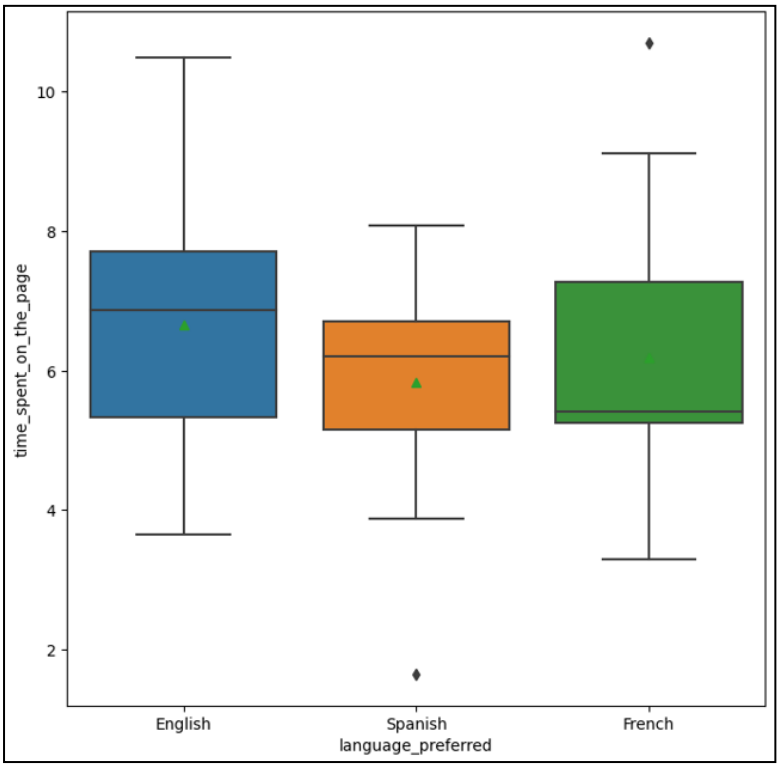
The p-value is 0.21298887487543447

Step 6: Compare the p-value with α.
As the p-value 0.21298887487543447 is greater than the level of significance, we fail to reject the null hypothesis.

Step 7: Draw inference: As the p-value is greater than 0.05, we fail to reject the null hypothesis. This means we don't have enough evidence to conclude that there is a significant association between converted status and preferred language.

Question 4: Is the time spent on the new page same for the different language users?

```
language_preferred
English    6.663750
French     6.196471
Spanish    5.835294
Name: time_spent_on_the_page, dtype: float64
```



Step 1: Define the null and alternate hypotheses:
*H*0: μlanguage1=μlanguage2=μlanguage3

Ha: At least one pair of means is different

Step 2: Select Appropriate test: This is a problem, concerning three population means.

Step 3: Decide the significance level
As given in the problem statement, we select α=0.05.

Step 4: Collect and prepare data: English, French and Spanish respectively.

```
1     7.13
16    8.73
22    3.65
23    7.02
25    4.39
Name: time_spent_on_the_page, dtype: float64
6      5.25
8     10.71
13     6.03
28     7.81
46     5.37
Name: time_spent_on_the_page, dtype: float64
2     4.40
4     4.75
12    5.86
15    6.27
24    6.18
Name: time_spent_on_the_page, dtype: float64
```

Step 5: Calculate the p-value: The p-value is 0.43204138694325955

Step 6: Compare the p-value with α.
As the p-value 0.43204138694325955 is greater than the level of significance, we fail to reject the null hypothesis.

Step 7: Draw inference:
As the p-value is greater than 0.05, we fail to reject the null hypothesis. This means we don't have enough evidence to conclude that the mean time spent on the new page is different for any pair of language groups.